5 Electromagnetic Waves

INTRODUCTION

Electromagnetic (EM) waves are composed of oscillating magnetic and electric fields. In general, EM waves are used for transporting energy and information. The typical examples of EM waves includes radio waves, TV signals, radar beams, light rays, etc.

Grey Matter Alert!

The speed of the electromagnetic wave is equal to the speed of light in a vacuum, and it changes depending on the material medium.

Characteristics of EM waves:

i) EM waves are propagated by oscillating electric and magnetic fields oscillating at right angles to each other.

ii) EM waves are not reflected by an electric or magnetic field.

iii) EM waves can show reflection and refraction, just like the light waves.

In this chapter, our major aim is to solve Maxwell's equations and derive EM wave motion in different given media:

a) Free space ($\sigma = 0$, $\in = \in_0$, $\mu = \mu_0$)

b) Loss-less dielectrics ($\sigma = 0$, $\in = \in_r \in_0$, $\mu = \mu_r \mu_0$)

c) Lossy dielectrics ($\sigma \neq 0$, $\in = \in_r \in_0$, $\mu = \mu_r \mu_0$)

d) Good conductor ($\sigma >>> 0$, $\in = \in_0$, $\mu = \mu_r \mu_0$)

MAXWELL'S EQUATIONS

AC analysis:

In the previous chapters, we mainly studied static fields, e.g. the electrostatic field denoted by E(x, y, z) and magnetostatic fields denoted by H(x, y, z). In static EM fields, electric and magnetic fields are independent of each other, whereas in dynamic EM fields, both fields are inter-dependent. In other words, a time varying electric field necessarily involves a corresponding time varying magnetic field. Time-varying electric and magnetic fields are represented by E(x, y, z, t) and H(x, y, z, t), respectively.

In a static field, the electric and magnetic fields are functions of space only while in time-varying field, the electric and magnetic fields are functions of space and time both.

Electrostatic fields are usually produced by static electric charges, whereas magnetostatic fields are due to the motion of electric charges with uniform velocity (direct current) or static magnetic charges. Time-varying fields or waves are usually due to time-varying currents or accelerated charges.

FARADAY's LAW AND INDUCED EMF

The electromotive force (EMF) or potential difference can be induced in any closed loop when the magnetic flux changes with time. In other words, a time varying field produces an induced voltage (EMF) in a closed circuit which causes a flow of current.

The induced EMF in any closed circuit is equal to the time rate of change of the magnetic flux linkage within the circuit, according to Faraday's law.

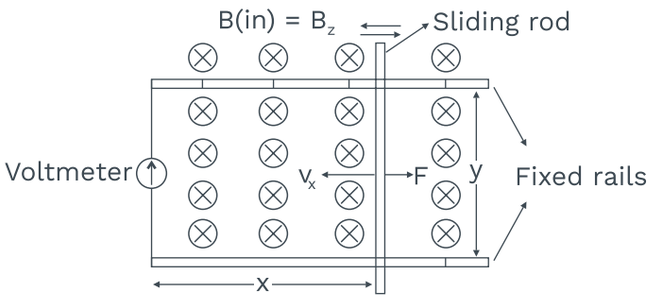


Fig. 5.1 Induced EMF Due to Sliding Rod

In the above example, a rod is sliding over a pair of fixed rails. Let the rod be moving with velocity v_x in the x-direction, and the magnetic field is in the z-direction. Therefore, both B and v are perpendiculars.

Now, the force on moving rod is:

$$F_y = Q(v_x \times B_z) + QE_y = 0$$

Initial induced EMF:

$$Q(v_x \times B_z) + QE_y = 0$$

$$\frac{dx}{dt} \cdot B = -E = -\frac{V}{y}$$

$$V = -yB\frac{dx}{dt}$$

If y and B are variable:

$$V = -\frac{d}{dt}(y \cdot B \cdot x)$$

$$V = -\frac{d}{dt}(B \cdot A) = -\frac{d}{dt}\psi_m \qquad (\because x \cdot y = A)$$

$$V_{emf} = -\frac{d\psi_m}{dt}$$

$$V_{emf} = \oint \vec{E} \cdot d\vec{l}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\psi_m}{dt} \qquad \ldots(i)$$

where $\vec{E}$ is the electric field and ψ is the flux through the closed loop. The negative sign shows that the induced voltage acts in such a way that it opposes the flux produced. This is known as Lenz's law.

Previous Years' Question

A loop is rotating about the y-axis in a magnetic field:

$$\vec{B} = B_0 \cos(\omega t + \phi)\hat{a}_x \text{ T.}$$

The voltage in the loop is:

a) zero
b) due to rotation only
c) due to transformer action only
d) due to both rotation and transformer action

$$\psi = \oint_s \vec{B} \cdot d\vec{s} = \iint \vec{B} \cdot d\vec{s}$$

Substitute the values in equation i):

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint \vec{B} \cdot d\vec{s}$$

By using Stokes's theorem:

$$\iint (\nabla \times \vec{E}) \cdot d\vec{s} = -\iint \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow \text{Faraday's law}$$

This is one of Maxwell's equations of time varying field.

Example 1:

A time varying magnetic flux linking a coil is given by $\psi = \frac{1}{4}\alpha t^3$. If at time t = 1s, the emf induced is 3V. Calculate the value of α.

Sol:

Given, $\psi = \frac{1}{4}\alpha t^3$

we know that,

$$V_{emf} = -\frac{d\psi}{dt} = -\frac{d}{dt}\left(\frac{1}{4}\alpha t^3\right) \quad ...(i)$$

$$V_{emf} = -\frac{1}{4} \times 3 \times \alpha \times t^2$$

$$V_{emf} = -\frac{3}{4}\alpha t^2 \quad ...(ii)$$

$$V_{emf}\big|_{t=1s} = 3 \ (\text{given})$$

$$3 = -\frac{3}{4}\alpha(1)^2$$

$$\alpha = -4 \text{ Wb/s}^2$$

Hence, the value of $\alpha = -4$ Wb/s^2

Rack Your Brain

Under what circumstances might the wires of the circuit radiate electro-magnetic waves if the current only flows in the direction?

Example 2:
The flux through each turn of 50 turn coil is $(t^2 + 2t)$ mWb, where t is in seconds. Find the value of induced emf at t = 3s.

Sol:
We know that,

$$\text{Induced emf} = -N\frac{d\psi}{dt}$$

where

N = Number of turns
ψ = Flux through each turn

$$\text{Induced emf} = -N\frac{d}{dt}(t^2 + 2t)\Big|_{t=3}$$

$$= -50[2t + 2]_{t=3}$$
$$= -50 \times 8$$
$$= -400 \text{ mV} = -0.4 \text{ V}$$

Hence, Induced emf in the coil is −0.4 V.

Rack Your Brain

Explain how the displacement current maintains the continuity of current in a circuit containing a capacitor.

Displacement current:
Displacement current is not a current due to the motion of charge carriers, but it is created when an external electric field is applied across a dielectric. Let us assume an example of a parallel plate capacitor connected across a battery, as shown in the below figure.

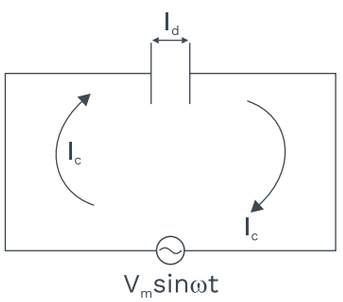


Fig. 5.2 Displacement Current

In the above circuit, I_c represents conduction current, and I_d represents displacement current. As the capacitor charges or discharges, a current flows through the circuit, but there is an insulated region between the plates of the capacitor; therefore, the actual charge transfer between the plates of the capacitor does not take place. Therefore, to complete the circuit, there is a displacement current between the plates of the capacitor, which is produced due to changing the electric flux.

Displacement current, $I_d = \frac{\in A}{d}\frac{dV}{dt}$

Displacement current density, $J_d = \frac{\partial D}{\partial t} = \epsilon_0 \frac{\partial E}{dt}$

In general, conduction current is very large in comparison of displacement current.

That is, $J_c >> J_d$

Now, for static EM fields, Maxwell's curl equation for magnetic fields:

$$\nabla \times \vec{H} = \vec{J} \qquad \ldots(1)$$

As the continuity equation of current is given by:

$$\nabla \cdot J = -\frac{\partial \rho_v}{\partial t} \qquad \ldots(2)$$

But from equation (1),

$$\nabla \cdot \vec{J} = \nabla \cdot (\nabla \times \vec{H}) = 0 \qquad \text{(Divergence of curl is zero)}$$

Therefore, equation (1) is not compatible with the time varying magnetic field. We must modify equation (1) to satisfy the continuity equation. Thus, the total current density is taken in place of the current density.

$$\vec{J}_t = \vec{J} + \vec{J}_d$$

Here,

$$\nabla \times \vec{H} = \vec{J} + \vec{J}_d$$

Now taking divergence:

$$\nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d$$

$$= \frac{-\partial \rho_v}{\partial t} + \nabla \cdot \vec{J}_d$$

as $\rho_v = \nabla \cdot \vec{D}$

$$\nabla \cdot \vec{J}_d = \frac{\partial}{\partial t}(\nabla \cdot \vec{D})$$

$$\nabla \cdot \vec{J}_d = \nabla \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

Now, time varying field Maxwell's equation becomes:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

where displacement current density is $\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$ and the conduction current density is $\vec{J} = \sigma \vec{E}$.

Example 3:
A parallel plate capacitor with a plate area of 6 cm^2 and plate separation of 1.5 mm has a voltage 30sin(100t)V applied to its plate. Calculate the displacement current assuming $\epsilon = 3\epsilon_0$.

Sol:
Given,

Area of plate, A = 6 cm^2
Plate separation, d = 1.5 mm
$\epsilon = 3\epsilon_0$
Voltage, V = 30 sin (100t)V

$$I_d = \frac{\epsilon A}{d}\frac{dV}{dt}$$

$$I_d = \frac{3 \times 8.85 \times 10^{-12} \times 6 \times 10^{-4}}{1.5 \times 10^{-3}}\frac{dV}{dt}$$

$$\frac{dV}{dt} = 30 \times 100 \cos 100t$$

$I_d = 12 \times 8.85 \times 10^{-13} \times 30 \times 100 \cos 100t$
$I_d = 31.86 \cos 100t$ nA

Rack Your Brain

How does the intensity of an electromagnetic wave depends on its electric field? How does it depend on its magnetic field?

Example 4:
A certain medium has a conductivity of 10^{-2}S/m and $\epsilon_r = 2$. Find conduction current density and displacement current density. If the electric field E = $5\times10^{-4}\sin10^{10}$t V/m.

Sol:
Given:
Conductivity, $\sigma = 10^{-2}$S/m

$\epsilon_r = 2$
$E = 5 \times 10^{-4}\sin10^{10}$t V/m

Conduction current density, $J_c = \sigma E$
$= 10^{-2} \times 5 \times 10^{-4}\sin10^{10}$t
$J_c = 5 \times 10^{-6} \sin10^{10}$t A/$m^2$

Displacement current density,

$$J_d = \epsilon \frac{dE}{dt}$$

$$J_d = \epsilon_0\epsilon_r \frac{d}{dt}\left(5 \times 10^{-4} \sin 10^{10}t\right)$$

$$= 2 \times 8.854 \times 10^{-12} \times 5 \times 10^{-4} \times 10^{10} \cos 10^{10}t$$

$J_d = 88.54 \times 10^{-16} \cos10^{10}$ t A/m^2

Maxwell's equations (final forms):

In Maxwell's equations, four equations all together provide complete information on the production and interrelation of electric and magnetic fields. These four equations represent four different experimental laws.

These four laws state that:

1) Electric field diverges from electric charge, an expression of Coulomb force:

$$\nabla \cdot \vec{D} = \rho_v$$

2) There are no isolated magnetic poles, but the Coulomb force acts between the poles of the magnet:

$$\nabla \cdot \vec{B} = 0$$

3) Electric fields are produced by changing magnetic fields, an expression of Faraday's law of induction:

That is, $$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4) Circulating magnetic fields are produced by changing electric fields and by electric current. This is the modification of Ampere's law to include the interaction of changing fields:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

The more generalised form of all these equations is as follows:

Differential form	Integral form	Experimental law
$\nabla \cdot \vec{D} = \rho_v$	$\oint_s \vec{D} \cdot d\vec{s} = \int_v \rho_v dV$	Gauss's law
$\nabla \cdot \vec{B} = 0$	$\oint_s \vec{B} \cdot d\vec{s} = 0$	Non-existence of isolated magnetic poles
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_L \vec{E} \cdot d\vec{l} = \frac{-\partial}{\partial t} \int_s \vec{B} \cdot d\vec{s}$	Faraday's law
$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint_L \vec{H} \cdot d\vec{l} = \int_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$	Ampere's circuit law

Table 5.1 Generalised form of All Equations

HARMONIC AND WAVES

Time harmonic field: A time harmonic field is one that periodically or sinusoidally varies with time.

EM wave: It is an electromagnetic field due to a time harmonic source such that it carries power.

For a clear understanding of EM wave propagation, we must know what waves are in general. A wave is a function of both space and time.

Helmholtz's equations:

Helmholtz's equation is a partial differential equation in the form of:

$$\nabla^2 A + k^2 A = 0$$

where

∇^2 = Laplacian operator

A = Harmonic function

k = Eigen value

Generally, the value of k is positive.

Grey Matter Alert!

Entire electromagnetic spectrum consist of radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays.

Helmholtz's equation for EM wave:

Consider a source with an electric field, $\vec{E}_s = E_0 e^{j\omega t}$ and magnetic field, $\vec{H}_s = H_0 e^{j\omega t}$

Now, from Maxwell's equation:

$$\nabla \times \vec{H} = \sigma \vec{E} + \in j\omega \vec{E} \qquad ...(1)$$

$$\left\{ \because \frac{d}{dt} = j\omega \right\}$$

And

$$\nabla \times \vec{E} = -j\omega\mu \vec{H}_s$$

$$\vec{H}_s = -\frac{1}{j\omega\mu}\left(\nabla \times \vec{E}\right)$$

$$\vec{H}_s = \left(-\frac{\nabla \times \vec{E}}{j\omega\mu}\right) \qquad ...(2)$$

Substituting the value of equation 2) in equation 1) and we get:

$$\nabla \times \left(\frac{\nabla \times \vec{E}}{-j\omega\mu} \right) = (\sigma + j\omega \in)\vec{E}$$

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2\vec{E} = -j\omega\mu(\sigma + j\omega \in)\vec{E}$$

As $\rho_v = 0$

$$\nabla \cdot \vec{D} = 0$$

Hence,

$$\nabla \cdot \vec{E} = 0$$

$$\nabla^2\vec{E}_s = j\omega\mu(\sigma + j\omega \in)\vec{E}_s \quad ...(3)$$

Similarly,

$$\nabla^2\vec{H}_s = j\omega\mu(\sigma + j\omega \in)\vec{H}_s \quad ...(4)$$

Equations 3) and 4) are Helmholtz's equation.

For EM-waves.

Let, $j\omega\mu(\sigma + j\omega \in) = \gamma^2$

Now, $\nabla^2\vec{E}_s - \gamma^2\vec{E}_s = 0$

$$\nabla^2\vec{H}_s - \gamma^2\vec{H}_s = 0$$

Let, $\vec{E}_s$ is x-directed, i.e. $\vec{E}_x$ and

$\vec{H}_s$ is x-directed, i.e. $\vec{H}_x$

Now, ∇ is z-directed, i.e. ∇_z

$$\frac{d^2\vec{E}_x}{\partial z^2} = \gamma^2\vec{E}_x \quad ...(5)$$

$$\frac{d^2\vec{H}_y}{\partial z^2} = \gamma^2\vec{H}_y \quad ...(6)$$

Here, the complex constant γ is defined as propagation constant.

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega \in)} = \alpha + j\beta$$

The real part of the propagation constant (α) is defined as the attenuation constant while the imaginary part (β) is defined as the phase constant.

Now, if we solve differential equations, 5) and 6) we get:

Previous Years' Question

If a plane electromagnetic wave satisfies the equation $\frac{\partial^2 E_x}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 E_x}{\partial t^2}$,

the wave propagates in the

a) x-direction
b) z-direction
c) y-direction
d) xy plane at an angle of 45° between the x- and z-directions

$$E(z)_x = C_1 e^{-\gamma z}\,\hat{a}_x \text{ or } C_2 e^{\gamma z}\,\hat{a}_x$$

$$H(z)_y = \underbrace{C_3 e^{-\gamma z}\,\hat{a}_y}_{\downarrow} \text{ or } \underbrace{C_4 e^{\gamma z}\,\hat{a}_y}_{\downarrow}$$

$+z$ direction $\quad -z$ direction

$$C_1 = E(\text{at } z = 0) = E_s = E_0 e^{j\omega t}$$

$$C_3 = H(\text{at } z = 0) = H_s = H_0 e^{j\omega t}$$

Now, wave equations become:

$$E(z,t)_x = E_0\, e^{j\omega t}\, e^{-\gamma z}\,\hat{a}_x$$

$$H(z,t)_y = \underbrace{H_0 e^{j\omega t}}_{\text{time}}\,\underbrace{e^{-\gamma z}\,\hat{a}_y}_{\text{space}}$$

Every EM wave is a time & space harmonic field advancement with transverse (orthogonal) $\vec{E}$ and $\vec{H}$ with propagation.

As $\gamma = \alpha + j\beta$

$$E(z,t)_x = E_0 e^{j\omega t} e^{-(\alpha+j\beta)z}\hat{a}_x$$

$$E(z,t)_x = E_0 e^{-\alpha z} e^{j(\omega t-\beta z)}\hat{a}_x$$

Similarly

$$H(z,t)_y = \underbrace{H_0 e^{-\alpha z}}_{\text{Amplitude}}\,\underbrace{e^{j(\omega t-\beta z)}\hat{a}_y}_{\text{Phase}}$$

Intrinsic impedance (η or Ω)

The intrinsic impedance describes the ratio of the electric field to the magnetic fields present in any medium. Intrinsic impedance of any uniform medium is dependent on the permittivity and permeability of the medium. When the conductivity of the medium varies, the intrinsic impedance also changes.

$$\eta = \frac{E_x}{H_y} = \frac{\gamma}{\sigma + j\omega \in} \quad (\Omega)$$

$$\eta = \sqrt{\frac{j\omega\mu}{(\sigma + j\omega \in)}} \quad \text{or} \quad \eta = \frac{-E_y}{H_x}$$

Rack Your Brain

If the electric and magnetic fields in a sinusoidal plane, wave were interchanged, in which direction relative to before would the energy propagates?

- Magnitude of intrinsic impedance indicate the magnitude ratio of electric field to the magnetic field.
- Angle of intrinsic impedance indicate the phase difference between electric and magnetic field.

Intrinsic impedance of free space:

For free space:

$\sigma = 0$

$\mu = \mu_0$

$\epsilon = \epsilon_0$

$$\therefore \quad \eta_0 = \sqrt{\frac{j\omega\mu_0}{0 + j\omega\epsilon_0}} = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$\eta_0 = \sqrt{\frac{4\pi \times 10^{-7}}{1/36\pi \times 10^{-9}}} = 120\pi \simeq 377\,\Omega$$

Intrinsic Impedance for free space = 377 Ω

Intrinsic impedance for dielectrics:

$\sigma = 0$

$\mu = \mu_0\mu_r$

$\epsilon = \epsilon_0\epsilon_r$

$$\eta = \sqrt{\frac{j\omega\mu}{0 + j\omega\epsilon}} = \sqrt{\frac{j\omega\mu_0\mu_r}{0 + j\omega\epsilon_0\epsilon_r}}$$

$$\eta = \sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}}$$

$$\therefore \quad \eta = 120\pi\sqrt{\frac{\mu_r}{\epsilon_r}} \quad (\Omega)$$

MATERIAL PROPAGATION

In this section, we are going to study about wave propagation in different mediums, e.g. loss-less medium, low-loss medium, high-loss medium, and changes in the characteristics of waves in different mediums.

Previous Years' Question

A material is described by the following electrical parameters at a frequency of 10 GHz.

$\sigma = 10^6$ mho/m, $\mu = \mu_0$ and $\epsilon/\epsilon_0 = -10$. The material at this frequency is considered to be assumed

$$\left(\epsilon_0 = -\frac{1}{36\pi} \times 10^{-9}\ \text{F/m}\right)$$

a) a good conductor
b) a good dielectric
c) neither a good conductor nor a good dielectric
d) a good magnetic material

Characterisation of medium:
Characterisation of medium is done on the basis of loss in power, during propagation in different mediums. There are three types of mediums:

1) No-loss or loss-less medium
2) Low-loss medium
3) High-loss medium

Loss-tangent (tan δ)
Loss-tangent is the measure of signal loss due to the inherent dissipation of electromagnetic energy in different mediums.

Loss tangent is the ratio of the magnitude of the conduction current density to the magnitude of the displacement current density.

Loss tangent,

$$\therefore \quad \tan\delta = \frac{|\vec{J}_c|}{|\vec{J}_d|}$$

Since,

$$\vec{J}_c = \sigma\vec{E}$$

$$\vec{J}_d = j\omega \in \vec{E}$$

$$\text{Loss tangent} = \frac{|\vec{J}_c|}{|\vec{J}_d|} = \frac{\sigma}{\omega \in}$$

Loss tangent = tanδ
where δ is loss angle

$$\therefore \quad \text{Loss angle, } \delta = \tan^{-1}\left(\frac{\sigma}{\omega \in}\right)$$

Now, the medium can be characterised on the basis of loss tangent value.

Case I:

If $\left(\frac{\sigma}{\omega \in}\right) = 0 \rightarrow$ Loss-less medium

$\left(\frac{\sigma}{\omega \in}\right) \neq 0 \rightarrow$ Lossy medium

Rack Your Brain

Discuss about the phase of the electric field and the magnetic field in loss-less medium.

Case II:

If $\left(\frac{\sigma}{\omega \in}\right) < 1 \rightarrow$ Low-loss medium

$\left(\frac{\sigma}{\omega \in}\right) >> 1 \rightarrow$ High-loss medium

1) Wave propagation is loss-less media

For loss-less media:

($\sigma = 0$, $\mu = \mu_r\mu_0, \in = \in_0\in_r$)

- Propagation constant

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega \in)} = \alpha + j\beta$$

Since, $\sigma = 0$

$$\therefore \quad \gamma = \sqrt{-\omega^2\mu \in} = j\omega\sqrt{\mu \in} = \alpha + j\beta$$

- Attenuation constant, $\alpha = 0$
- Phase constant, $\beta = \omega\sqrt{\mu \in}$
- Velocity = $\frac{\omega}{\beta}$
- Intrinsic impedance:

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega \in}}$$

$$\eta = \sqrt{\frac{\mu}{\in}} \text{ (real)} \qquad (\therefore \sigma = 0)$$

2) Wave propagation in lossy-medium

($\sigma \neq 0$, $\mu = \mu_0\mu_r$, $\in = \in_0\in_r$)

- Propagation constant,

$$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega \in)}$$

$$\alpha^2 - \beta^2 + 2j\alpha\beta = j\omega\mu\sigma - \omega^2\mu\in$$

After solving this:

$$\alpha = \omega\sqrt{\frac{\mu \in}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega \in}\right)^2} - 1\right)}$$

$$\beta = \omega\sqrt{\frac{\mu \in}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega \in}\right)^2} + 1\right)}$$

a) Low-loss medium $\left(\frac{\sigma}{\omega \in} << 1\right)$

Attenuation constant, $\alpha = \frac{\sigma}{2}\sqrt{\frac{\mu}{\in}}$

Phase constant, $\beta = \omega\sqrt{\mu \in}$

Velocity, $v = \frac{1}{\sqrt{\mu \in}}$

Impedance, $\eta = \sqrt{\frac{\mu}{\in}}$ (real)

b) High-loss medium $\left(\frac{\sigma}{\omega \in} >> 1\right)$ or $(\sigma >> \omega\in)$

All the good conductors are high-loss medium. In good conductor, displacement current is negligible in comparison to conduction current.

$\therefore \quad J_{total} = \sigma E + j\omega \in E$

As $\sigma >> \omega\in$, $|J_{conduction}| >> |J_{displacement}|$

- Propagation constant

$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega \in)}$

$\because \quad \sigma >> \omega\in$

$\gamma = \sqrt{j\omega\mu\sigma} = \sqrt{\omega\mu\sigma \angle 90°}$

$\alpha + j\beta = \sqrt{\omega\mu\sigma} \angle 45°$

$\alpha + j\beta = \sqrt{\omega\mu\sigma}(\cos 45° + j \sin 45°)$

$\gamma = \alpha + j\beta = \sqrt{\frac{\omega\mu\sigma}{2}} + j\sqrt{\frac{\omega\mu\sigma}{2}}$

- Attenuation constant, $\alpha = \sqrt{\frac{\omega\mu\sigma}{2}}$
- Phase constant, $\beta = \sqrt{\frac{\omega\mu\sigma}{2}}$

Attenuation in high-loss medium increases with an increase in the frequency and conductivity, and decreases with an increase in wavelength.

Previous Years' Question

The intrinsic impedance of a lossy dielectric medium is given by:

a) $\frac{j\omega\mu}{\sigma}$ **b)** $\frac{j\omega \in}{\mu}$

c) $\sqrt{\frac{j\omega\mu}{(\sigma + j\omega \in)}}$ **d)** $\sqrt{\frac{\mu}{\in}}$

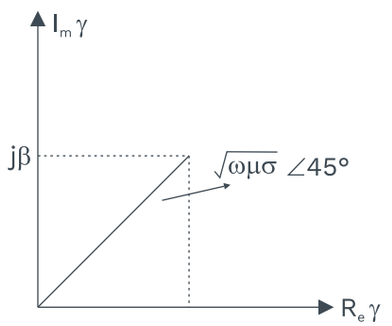


Fig. 5.3 Propagation Constant in High-Loss Medium (or Good Conductor)

- Intrinsic impedance

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega \in}}$$

$\because \quad \sigma >> \omega\in$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}\angle 90°}$$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}}\angle 45°$$

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}}(\cos 45° + j\sin 45°)$$

$$\eta = \sqrt{\frac{\omega\mu}{2\sigma}} + j\sqrt{\frac{\omega\mu}{2\sigma}}$$

$$\eta = R + jX = \sqrt{\frac{\omega\mu}{2\sigma}} + j\sqrt{\frac{\omega\mu}{2\sigma}} \quad (\Omega)$$

Example 5:
The electric field components of an EM wave propagating through a medium (characterized by $\in = 3\in_0$, $\mu = 6\mu_0$, and $\sigma = 0.003$ S/m) is

$$\vec{E}(z, t) = 9e^{-\alpha z}\cos(2\pi 30 \times 10^5 t - \beta z)\hat{x}\ V/m$$

Calculate the value of:

a) Propagation constant, γ
b) Attenuation constant α
c) Phase constant, β

Sol:

Given:

$\omega = 2\pi \times 30 \times 10^5$

$f = 30 \times 10^5 Hz = 3MHz$

$\in = 3\in_0$

$\mu = 6\mu_0$

$\sigma = 0.03$ S/m

a) Propagation constant,

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega \in)}$$

$$\gamma = \sqrt{j\omega\mu\sigma - \omega^2\mu \in}$$

$$\gamma = \sqrt{j(2\pi \times 30 \times 10^5 \times 6 \times 4\pi \times 10^{-7} \times 0.03) - (2\pi \times 30 \times 10^5)^2 6 \times 4\pi \times 10^{-7} \times 3 \times 8.854 \times 10^{-12}}$$

$$\gamma = \sqrt{4.26j - 0.07}$$

$$\gamma = \sqrt{4.26\angle 90.94} = 2.06\angle 45.47$$

or

$\gamma = r(\cos\theta + j\sin\theta)$

$= 2.06\cos 45.47 + 2.06j \sin 45.47$

$\gamma = 1.44 + j1.47$

Propagation constant, $\gamma = 1.44 + j1.47$

And we know that

$\gamma = \alpha + j\beta$

After comparing we get:

b) Attenuation constant $\alpha = 1.44$ Nepers/m

c) Phase constant, $\beta = 1.47$ rad/m

Note:

We can also find α and β by direct formulas.

$$\alpha = \omega\sqrt{\frac{\mu \in}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega \in}\right)^2} - 1\right]}$$

and,

$$\beta = \omega\sqrt{\frac{\mu \in}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega \in}\right)^2} + 1\right]}$$

Previous Years' Question

Copper behaves as a:

a) conductor always.

b) conductor or dielectric depending on the applied electric field strength.

c) conductor or dielectric depending on the frequency.

d) conductor or dielectric depending on the electric current density.

Example 6:

The EM propagation in a non-magnetic medium is

$$\vec{E} = 7\cos(2\pi \times 10^7 t + 7x)\hat{a}_z \text{ V/ m}$$

Find the:

a) Direction of wave propagation
b) Wavelength (λ)
c) Velocity (v)
d) Relative permittivity (ϵ_r)
e) Intrinsic impedance (η)

Sol:

Given:

$$\vec{E} = 7\cos(2\pi \times 10^7 t + 7x)\hat{a}_z \text{ V/ m}$$

a) Wave is travelling in negative x-direction.

b) From given wave:

$\omega = 2\pi \times 10^7$, $\beta = 7$

$$f = \frac{\omega}{2\pi} = 10^7 \text{Hz}$$

$$\beta = \frac{2\pi}{\lambda}$$

Wavelength, $\lambda = \frac{2\pi}{7} = 0.898\,\text{m}$

c) Velocity, $v = \frac{\omega}{\beta} = \frac{2\pi \times 10^7}{7}$

$$v = 8.98 \times 10^6 \text{ m/s}$$

d) Velocity $= \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$

$\{\because \quad \mu_r = 1$ (non-magnetic medium)$\}$

$$\text{Velocity} = \frac{c}{\sqrt{\epsilon_r}} \qquad \left(\because \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c\right)$$

$$v = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} = 8.98 \times 10^6$$

$$\sqrt{\epsilon_r} = \frac{3 \times 10^8}{8.98 \times 10^6} = 33.41$$

Relative permittivity, $\epsilon_r \simeq 1116$

e) Intrinsic impedance,

$$\eta = \sqrt{\frac{\mu}{\in}} = \frac{120\pi}{\sqrt{\in_r}}$$

$$\eta = \frac{120\pi}{33.41} = 11.28\ \Omega$$

Intrinsic impedance, $\eta = 11.28\ \Omega$

Example 7:

The electric field components of an EM wave propagating through a medium (characterized by $\in = 3\in_0$, $\mu = 6\mu_0$, and $\sigma = 0.03$ S/m) is

$$\vec{E}(z,t) = 10e^{-\alpha z}\cos(6\pi \times 10^6 t - \beta z)\hat{x}\ V/m$$

Find the magnetic field intensity vector $\vec{H}(z,t)$.

Sol:

We know that:

$$\frac{E_x}{H_y} = \eta$$

$$H_y = \frac{E_x}{\eta}$$

Intrinsic impedance $\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega \in}}$

$$\eta = \sqrt{\frac{j \times 2\pi \times 30 \times 10^5 \times 6 \times 4\pi \times 10^{-7}}{0.03 + j2\pi \times 30 \times 10^5 \times 3 \times 8.854 \times 10^{-12}}}$$

$$\eta = \sqrt{\frac{j(142.12)}{0.03 + 5 \times 10^{-4} j}} = 68.83\angle 44.5$$

$$\vec{E}(z,t) = 10e^{-\alpha z}\cos(6\pi \times 10^6 t - \beta z)\hat{x}\ V/m$$

$$H_y = \frac{E_x}{\eta}$$

Here, $E_x = 10$ (magnitude)

$$H_y = \frac{10}{68.83\angle 44.5} = 145.3 \times 10^{-3} \angle -44.5$$

$$H(z,t) = 145.33\, e^{-\alpha z}\cos(6\pi \times 10^6 t - \beta z - 44.5)\hat{y} \quad mA/m$$

Skin depth or depth of penetration (δ)

Skin depth is defined as the distance over which a plane wave is attenuated by a factor of e^{-1} in a lossy medium.

$|E| = E_0 e^{-\alpha z}$

$\Rightarrow$ If $z \to \infty$,

then $|E| = 0$

$\Rightarrow$ If $z \to \frac{1}{\alpha}$,

then $|E| = E_0 e^{-1} = 0.37E$

That is when a wave travels for $\frac{1}{\alpha}$ distance, its amplitude decays to $\frac{1}{e}$ times of the initial value.

This distance, $\frac{1}{\alpha} = \delta$ = skin depth

- For dielectric and loss-less medium:
  Skin depth $\to \infty$
- For conductor or lossy medium:
  Skin depth $\to 0$

That is skin depth is very small.

- For conductors:

$$\delta = \frac{1}{\alpha}$$

and $\alpha = \sqrt{\frac{\omega\mu\sigma}{2}}$

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$$

$$\delta = \frac{1}{\sigma}\sqrt{\frac{2\sigma}{\omega\mu}} = \frac{1}{\sigma R_s}$$

where R_s is a real part of the intrinsic impedance (η) of the medium.

Therefore,

$$\text{Skin depth}, \delta = \frac{1}{\alpha} = \frac{1}{\beta} = \sqrt{\frac{2}{\omega\mu\sigma}}$$

$$\delta = \frac{1}{\sigma R_s} = \frac{1}{\sigma X_s}$$

Grey Matter Alert!

Submarine communication, i.e. under water communication is done at very low frequency because skin depth is maximum for low frequency $\left(\text{as}, \delta \propto \frac{1}{\sqrt{f}}\right)$.

- The velocity of propagation within the good conductor in terms of skin depth:

  Velocity (v) $= \frac{\omega}{\beta} = \omega\delta$

$\therefore$ Velocity (v) = $\omega\delta$

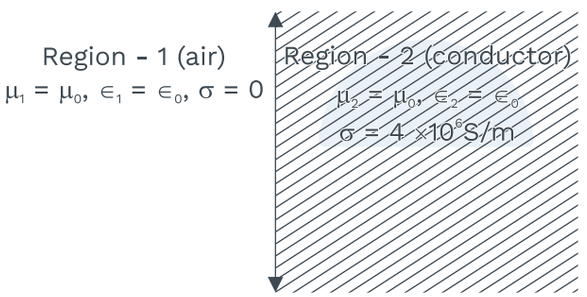
- The wavelength within the good conductor in terms of skin depth:

  $\lambda = \frac{2\pi}{\beta} = 2\pi\delta$

$\therefore$ $\lambda = 2\pi\delta$

Example 8:
Consider a uniform plane wave with f = 2 MHz at an air/conductor interface as shown below. Find the value of the following in region 2.

a) Velocity (v)
b) Skin depth (δ)
c) Attenuation constant (α) and phase constant (β)
d) wavelength (λ)



Sol:

a) Velocity, $v = \frac{\omega}{\beta}$

For conductors (high loss)

Velocity $= \sqrt{\frac{2\omega}{\mu\sigma}}$

Velocity $= \sqrt{\frac{2 \times 2\pi \times 2 \times 10^6}{4\pi \times 10^{-7} \times 4 \times 10^6}}$

Velocity, $v = \sqrt{\frac{1}{2 \times 10^{-7}}}$

Velocity, $v \simeq 2236\,m/s$

Previous Years' Question

The skin depth of copper at a frequency of 3 GHz is 1 micron (10^{-6} m). At 12 GHz, for a non-magnetic conductor whose conductivity is 1/9 times that of copper, the skin depth would be:

a) $\sqrt{9 \times 4}$ microns **b)** $\sqrt{\frac{9}{4}}$ microns

c) $\sqrt{\frac{4}{9}}$ microns **d)** $\frac{1}{\sqrt{9 \times 4}}$ microns

b) Skin depth (δ):

$\because \quad v = \frac{\omega}{\beta}$

$\delta = \frac{1}{\beta}$

$v = \omega\delta$

$\therefore \quad \delta = \frac{v}{\omega} = \frac{2236}{2\pi \times 2 \times 10^6} = 177.94\,\mu m$

c) Attenuation constant,

$\alpha = \frac{1}{\delta} = \frac{1}{177.94 \times 10^{-6}}$

$\therefore \quad \alpha = 5.62 \times 10^3$ Np/m

Phase constant,

$\beta = \alpha = \frac{1}{\delta} = 5.62 \times 10^3$ rad/m

d) Wavelength,

$\lambda = \frac{v}{f} = \frac{2236}{2 \times 10^6} = 1118 \times 10^{-6} m$

$\lambda = 1118\ \mu m$

Rack Your Brain

Why is the power cable very thin and multiple wires are used? Why can't we use a single thick wire?

Example 9:

Skin depth for an EM wave in a high-loss medium 'A' is 3.2 μm. Calculate the skin depth for medium B.

(Given, $\sigma_A = 3 \times 10^4$, $\sigma_B = 5 \times 10^4$, $\epsilon_1 = \epsilon_2$, $\mu_1 = \mu_2$)

Sol:

Skin depth $\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}}$

As all parameters are the same except σ

$\Rightarrow \delta \propto \frac{1}{\sqrt{\sigma}}$

$\Rightarrow \frac{\delta_A}{\delta_B} = \sqrt{\frac{\sigma_B}{\sigma_A}}$

$\Rightarrow \frac{\delta_A}{\delta_B} = \sqrt{\frac{5 \times 10^4}{3 \times 10^4}}$

$\Rightarrow \delta_B = \frac{\delta_A}{1.29}$

$$\delta_B = \frac{3.2 \times 10^{-6}}{1.29} = 2.48\,\mu m$$

Skin depth for medium B, δ_B = 2.48 μm

Power and power density:

Power is the amount of energy transferred or converted per unit of time.

Power = V I

$$P = \frac{W}{Q} \cdot \frac{Q}{t}$$

$$P = \frac{W}{t} = \frac{Energy}{time}$$

$$Power = \frac{Energy}{time}$$

In network theory, we calculate power dissipation in components, which is the product of voltage (V) and current (I). But in waves and 3D spread of fields, we discuss about power density or strength of power.

Power density is the amount of power per unit volume.

Power density is calculated using power flow in the EM field. It can be obtained using Maxwell's equations.

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \qquad ...(1)$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \in \frac{\partial \vec{E}}{\partial t} \qquad ...(2)$$

And multiply $\vec{E}$ on both sides in equation 2),

$$(\nabla \times \vec{H}) \cdot \vec{E} = \sigma \vec{E}^2 + \in \frac{\partial \vec{E}}{\partial t} \vec{E}$$

$$\vec{H}(\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H}) = \sigma \vec{E}^2 + \in \frac{\partial \vec{E}}{\partial t} \vec{E}$$

$$-\nabla \cdot (\vec{E} \times \vec{H}) - \sigma \vec{E}^2 = \in \frac{\partial \vec{E}}{\partial t} \vec{E} + \mu \frac{\partial \vec{H}}{\partial t} \cdot \vec{H}$$

As, $\frac{\partial \vec{E}}{\partial t} \cdot \vec{E} = \frac{\partial}{\partial t} \frac{\vec{E}^2}{2}$ $\left\{ \because x dx = d\left(\frac{x^2}{2}\right) \right\}$

$$\nabla \cdot (\vec{E} \times \vec{H}) + \sigma \vec{E}^2 = -\left(\in \frac{d}{dt}\left(\frac{E^2}{2}\right) + \mu \frac{d}{dt}\left(\frac{H^2}{2}\right) \right)$$

By taking volume integral on both sides:

$$\oiint_s (\vec{E}\times\vec{H})\,ds + \iiint \sigma\vec{E}^2 dV = -\frac{d}{dt}\iiint\left[\frac{1}{2}\in E^2 + \frac{1}{2}\mu H^2\right]dV$$

$$\oiint_s (\vec{E}\times\vec{H})ds = -\frac{\partial}{\partial t}\iiint\left[\frac{1}{2}\in E^2 + \frac{1}{2}\mu H^2\right]dV - \iiint \sigma\vec{E}^2\,dV$$

Here, $\oiint_s (\vec{E}\times\vec{H})\,ds$ = Total power

$\frac{1}{2}\in\vec{E}^2 \rightarrow$ Energy density in $\vec{E}$ field

$\frac{d}{dt}\frac{1}{2}\in\vec{E}^2 \rightarrow$ Power density in $\vec{E}$ field

$\iiint\frac{d}{dt}\frac{1}{2}\in\vec{E}^2 dV \rightarrow$ Power in $\vec{E}$ field

Similarly

$\iiint\frac{d}{dt}\frac{1}{2}\mu\vec{H}^2 dV \dashrightarrow$ Power in $\vec{H}$ field

$\iiint\sigma\vec{E}^2 dV \rightarrow$ Ohmic power dissipation

Previous Years' Question

The incoming solar radiation at a place on the surface of the earth is 1.2 kW/m^2. The amplitude of the electric field corresponding to this incident, power is nearly equal to:

a) 80 mV/m **b)** 2.5 V/m
c) 30 V/m **d)** 950 V/m

Poynting vector:

The quantity $\vec{E}\times\vec{H}$ is known as the poynting vector, measured in W/m^2.

$$\vec{P} = \vec{E}\times\vec{H}$$

Poynting vector $\vec{P}$ is along the direction of wave propagation.

Poynting theorem: The net power flowing out of any given volume V is equal to the time rate of decrease in the energy stored within the given volume minus the ohmic losses. The term $\vec{E}\times\vec{H}$ signifies the EM wave power density, which crosses the surface in the direction of propagation.

Time-average power density:

$$P_{avg} = E_{rms}\times H_{rms}$$

$$= \frac{E_0 e^{-\alpha z}}{\sqrt{2}}\times\frac{H_0 e^{-\alpha z}}{\sqrt{2}}$$

$$E(z,t)_x \times H(z,t)_y = P(z,t)_z$$

$$P(z)_{avg} = \frac{1}{2}E_0 H_0 e^{-2\alpha z} = P_0 e^{-2\alpha z}$$

where

$$P_0 = \frac{1}{2}E_0H_0 = P \qquad (\text{at } z = 0)$$

$$P(z)_{avg} = \frac{1}{2}\text{Re}(\vec{E} \times \vec{H})$$

If amplitude decays by α rate, then the power decays by 2α rate.

$$P(z)_{avg} = \frac{1}{2}(\vec{E} \times \vec{H})$$

The total time average power crossing a given surface s is given by:

$$P_{avg} = \int_s \vec{P}_{avg} \cdot d\vec{s}$$

Conclusion

1) Power is always amplitude dependent and independent of phase and type of the harmonic.

2) $\vec{P}_{avg}$ is a vector function of (x, y, z) with units W/m^2.

Example 10:

In free space, the electric field given by $\vec{E} = 30\sin(\omega t - \beta z)\hat{a}_y$ V/m. Find the average power crossing a circular area of radius 2 m in plane $z = \frac{1}{2}$.

Sol:

Given:

$$\vec{E} = 30\sin(\omega t - \beta z)\hat{a}_y$$

Average pointing vector, $\vec{P} = \vec{P}_{avg}$

$$\vec{P}_{avg} = \frac{1}{2\eta}|\vec{E}|^2 \hat{a}_z$$

$$|\vec{E}| = 30$$

$$\vec{P}_{avg} = \frac{1}{2 \times 120 \times \pi} \times (30)^2 \hat{a}_z$$

$$\vec{P}_{avg} = \frac{900}{240\pi}\hat{a}_z \quad W/m^2$$

As $\vec{P}_{avg}$ is normal to plane $z = \frac{1}{2}$, so average power crossing the area is given by:

Rack Your Brain

Discuss about the requirement of EM-wave for carrying power.

$$\text{Power} = \vec{P}_{avg} \times \text{Area}$$

$$= \frac{900}{240\pi} \times \pi(2)^2 = \frac{900}{60}\text{W} = 15\text{ W}$$

Average power crossing through the given area = 15 W.

WAVE POLARISATION

Polarisation is defined as electric field orientation and the possible planner components satisfying the basic transverse nature. It is the study of the time behaviour of EM Wave.

The polarisation of a uniform plane wave defines the locus drawn by the electric field vector as the wave propagates.

Linear polarisation

In linear polarisation the electric field of light is confined to a single plane along the direction of propagation. This is the phase difference between different components of electric field vector is $n\pi$, i.e. $\Delta\phi = \pm n\pi$, n = 0, 1, 2, ... $E(z,t) = E_x \sin(\omega t - \beta z)\hat{a}_x + E_y \sin(\omega t - \beta z)\hat{a}_y$ 1)

Here, E_x and E_y are the maximum magnitudes of the x and y components respectively and ϕ_x and ϕ_y are the initial phase differences of x and y components, respectively.

As the wave is propagating in positive z-direction:

$$\hat{a}_x \times \hat{a}_y = \hat{a}_z$$

$$\Delta\phi = \phi_y - \phi_x$$

In the given equation,

$$\Delta\phi = 0$$

Hence, it is an example of linear polarisation.

Example 11:

The polarisation of the wave whose electric field vector is given by:

$$\vec{E} = 2\sin(\omega t - \beta x)\hat{a}_y + 3\sin(\omega t - \beta x - \pi)\hat{a}_Z \text{ V/ m}$$

Sol:

Wave is propagating in positive x-direction:

$$\hat{a}_y \times \hat{a}_z = \hat{a}_x$$

$$= \phi_z - \phi_y$$

$$\Delta\phi = \pi$$

Previous Years' Question

The time averaged Poynting vector, in W/m^2, for a wave with $\vec{E} = 24e^{(\omega t+\beta z)}\vec{a}_y$ V/m in free space is

a) $-\frac{2.4}{\pi}\vec{a}_z$ **b)** $\frac{2.4}{\pi}\vec{a}_z$

c) $\frac{2.8}{\pi}\vec{a}_z$ **d)** $-\frac{2.8}{\pi}\vec{a}_z$

This satisfies the condition of linear polarisation.
Hence, the given wave is linearly polarised.

Circular polarisation

Circular polarisation is achieved when the two electric field components are having equal amplitude's and out of phase by

$$\pm(2n+1)\frac{\pi}{2},\ n = 0,\ 1,\ 2,\ 3,\ldots$$

- Let wave is propagating in positive z-direction and electric field components are in x- and y-direction. Then condition for circular polarisation is:
  $|\vec{E}_x| = |\vec{E}_y|$

($|\vec{E}_x|$ and $|\vec{E}_y|$ are maximum amplitude of x- and y-components, respectively.)

$$\Delta\phi = \phi_y - \phi_x = (2n+1)\frac{\pi}{2} \quad \left\{\begin{array}{c}\text{Left circularly}\\ \text{polarised}\\ \text{or clockwise}\end{array}\right\}$$

$$\Delta\phi = -(2n+1)\frac{\pi}{2} \quad \left\{\begin{array}{c}\text{Right circularly}\\ \text{polarised or}\\ \text{counter-clockwise}\end{array}\right\}$$

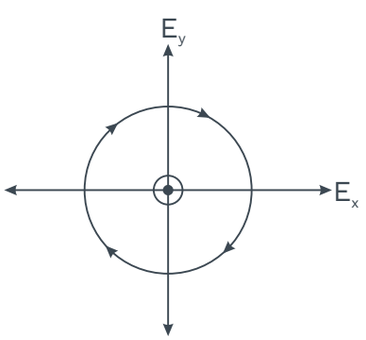


Fig. 5.4 Left Circularly Polarised (or) Clockwise

Rack Your Brain

What is the cause of polarisation of electromagnetic wave?

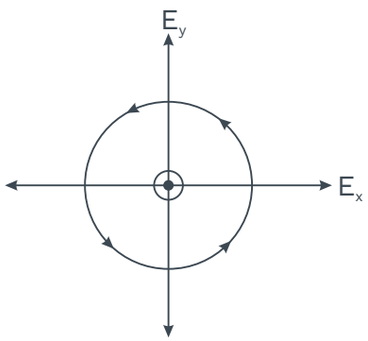


Fig. 5.5 Right Circularly Polarised (or) Counter Clockwise

Circular polarisation is preferred for antennas of unknown polarisation.

Elliptical polarisation:

In general, all the waves which are not linear or circular are elliptical.

Condition for elliptical polarisation:

$$|\vec{E}_x| \neq |\vec{E}_y|$$

$$\Delta\phi = \phi_y - \phi_x = (2n+1)\frac{\pi}{2} \qquad \left\{\begin{matrix}\text{Left elliptically} \\ \text{polarised}\end{matrix}\right\}$$

$$\Delta\phi = -(2n+1)\frac{\pi}{2} \qquad \left\{\begin{matrix}\text{Right elliptically} \\ \text{polarised}\end{matrix}\right\}$$

Depends also on the direction of the wave propagation

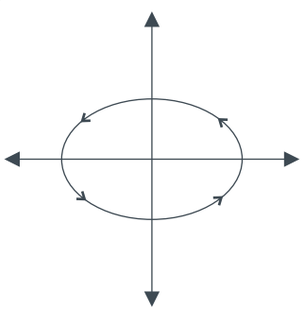


Fig. 5.6 Right Elliptically Polarised (or) Counter Clockwise

Previous Years' Question

A plane electromagnetic wave travelling along the +z-direction, has its electric field given by $E_x = 2\cos(\omega t)$ and $E_y = 2\cos(\omega t + 90°)$, the wave is:

a) linearly polarised
b) right circularly polarised
c) left circularly polarised
d) elliptically polarised

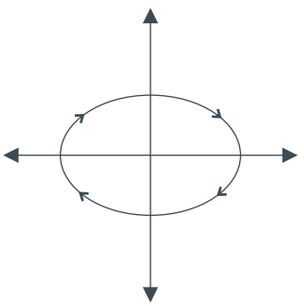


Fig. 5.7 Left Elliptically Polarised (or) Clockwise

Example 12:

The electric field vector of a plane wave is given by

$$\vec{E} = 2\cos\left(\omega t + \beta z - \frac{\pi}{2}\right)\hat{a}_x + \cos(\omega t + \beta z)\hat{a}_y \text{ V/ m}$$

what is the polarisation of waves?

Sol:

Given:

$$\vec{E} = 2\cos\left(\omega t + \beta z - \frac{\pi}{2}\right)\hat{a}_x + \cos(\omega t + \beta z)\hat{a}_y \text{ V/ m}$$

$$\left|\vec{E}_x\right| = 2$$

$$\left|\vec{E}_y\right| = 1$$

$$\left|\vec{E}_x\right| \neq \left|\vec{E}_y\right|$$

As the direction of the wave is negative z-direction.

$$\hat{a}_y \times \hat{a}_x = -\hat{a}_z$$

Hence,

$$\left.\begin{array}{l} \phi_x - \phi_y = -\frac{\pi}{2} \\ |\vec{E}_x| \neq |\vec{E}_y| \end{array}\right\} \text{ Right elliptical polarization}$$

and

Hence, the wave is right elliptically polarised.

Example 13:

The electric field vector of a plane wave is given by

$$\vec{E} = (2\hat{a}_y + 2j\hat{a}_z)\, e^{j(\omega t - \beta x)} \text{ V/ m}$$

Rack Your Brain

What are orthogonal sets of polarisation?

Previous Years' Question

The polarisation of a wave with electric field vector $\vec{E} = E_0\, e^{(\omega t - \beta z)}(\vec{a}_x + \vec{a}_y)$ is:

a) linear
b) elliptical
c) left-hand circular
d) right-hand circular

Sol:
Given:

$$\vec{E} = (2\hat{a}_y + 2j\hat{a}_z)e^{j(\omega t - \beta x)} \text{ V/ m}$$

$$\left|\vec{E}_y\right| = 2$$

$$\left|\vec{E}_z\right| = 2$$

Wave is propagating in positive x-direction:

$$\hat{a}_y \times \hat{a}_z = \hat{a}_x$$

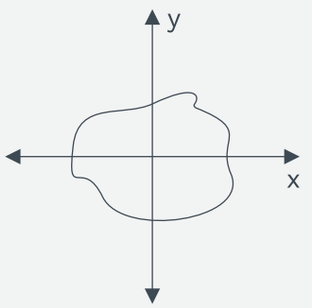
$$\left.\begin{aligned}\phi_z - \phi_y &= \frac{\pi}{2} \\ |\vec{E}_y| &= |\vec{E}_z|\end{aligned}\right\} \text{Left circular}$$

Hence, the given wave is left circularly polarised.

Grey Matter Alert!

General case: $\Delta\phi \neq 0, \quad \Delta\phi \neq \frac{\pi}{2}$

In general, EM wave is elliptically polarised.



Example 14:
The electric field vector of a plane wave is given by

$$\vec{E} = 3\cos(\omega t + \beta z - 60°)\hat{a}_x + 3\sin(\omega t + \beta z)\hat{a}_y \text{ V/ m}$$

Find the polarisation of the plane wave.

Sol:
Given:

$$\vec{E} = 3\cos(\omega t + \beta z - 60°)\hat{a}_x + 3\sin(\omega t + \beta z)\hat{a}_y \text{ V/ m}$$

$$\left|\vec{E}_x\right| = 3$$

$$\left|\vec{E}_y\right| = 3$$

$$\left|\vec{E}_x\right| = \left|\vec{E}_y\right|$$

$$\vec{E} = 3\cos(\omega t + \beta z - 60°)\hat{a}_x + 3\sin(\omega t + \beta z)\hat{a}_y \text{ V/ m}$$

$$\vec{E} = 3\cos(\omega t + \beta z - 60°)\hat{a}_x + 3\cos(\omega t + \beta z - 90°)\hat{a}_y \text{ V/ m}$$

As the wave is propagating in the negative z-direction:

$$\hat{a}_y \times \hat{a}_x = -\hat{a}_z$$

$$\left.\begin{aligned}\phi_x - \phi_y &= -60° + 90° = 30° \\ |\vec{E}_x| &= |\vec{E}_y|\end{aligned}\right\} \text{Left elliptically}$$

Hence, the wave is left elliptically polarised.

REFLECTION AND TRANSMISSION OF EM WAVE

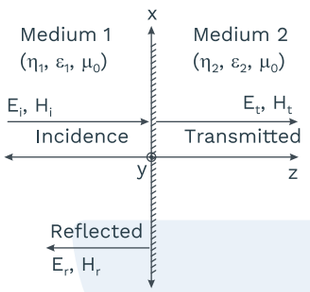


Fig. 5.8 Reflection and Transmission

When a plane wave propagating in a uniform medium faces a different medium, a portion of the wave is reflected from the interface and remaining wave is transmitted to other mediums. For finding reflected and transmitted waves, electromagnetic field boundary conditions at the medium interface are used.

The reflection coefficient is the ratio of the amplitude of the reflected wave to the incidence wave.

$$\left.\begin{aligned}\frac{E_r}{E_i} &= \Gamma_E \\ \frac{H_r}{H_i} &= \Gamma_H\end{aligned}\right\} \text{Reflection coefficient}$$

$$E_i = \eta_1 H_i$$
$$E_t = \eta_2 H_t$$
$$E_r = -\eta_1 H_r$$

When reflection takes place, it changes the direction of propagation and one of the fields has a phase shift of 180°.

For the normal incidence of wave on the boundary:

$$\left.\begin{aligned}E_{t1} &= E_{t2} \\ H_{t1} &= H_{t2}\end{aligned}\right\} \text{Tangential field}$$

$$E_{t1} = E_i + E_r = E_{t2}$$

Now, $E_i + E_r = E_t$ ...(i)

$H_i + H_r = H_t$ ...(ii)

And, $E_i = \eta_1 H_i$

$E_r = -\eta_1 H_r$

$E_t = \eta_2 H_t$

Divide equation i) by E_i

$$1 + \frac{E_r}{E_i} = \frac{E_t}{E_i}$$

$$1 + \Gamma = \tau \quad \text{...(iii)}$$

where $\frac{E_t}{E_i} = \tau$ transmission coefficient

The transmission coefficient is the ratio of the amplitude of the transmitted wave to incident wave.

Now, if we replace H in equation ii) by E, then we get:

$$\frac{E_i}{\eta_1} - \frac{E_r}{\eta_1} = \frac{E_t}{\eta_2}$$

$$E_i - E_r = \frac{\eta_1}{\eta_2} E_t \quad \text{...(iv)}$$

Now divide equation iv) by E_i

$$1 - \Gamma = \frac{\eta_1}{\eta_2} \tau \quad \text{...(v)}$$

Solving equations iii) and v):

$$\Gamma_E = \frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

and $\tau_E = \frac{E_t}{E_i} = 1 + \Gamma_E = \frac{2\eta_2}{\eta_2 + \eta_1}$

Now for magnetic field:

$$\Gamma_H = \frac{H_r}{H_i} = \frac{-E_r}{\eta_1 \frac{E_i}{\eta_1}}$$

$$\Gamma_H = -\Gamma_E = \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2}$$

Grey Matter Alert!

The transmitted waves never undergo any phase change for dielectric media interface. While reflected wave may undergoes a phase change.

Example 15:
An EM wave is an incident from air to a non-magnetic dielectric of $\varepsilon_r = 16$ (normal). Find the following:

a) The transmitted power fraction into the dielectric.

b) The reflected $\vec{E}$ field fraction.

c) Transmitted $\vec{H}$ field fraction.

Sol:

$$\eta_1 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi$$

$$\eta_2 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{120\pi}{4}$$

$$\Gamma_E = \frac{\eta_2 - \eta_1}{\eta_1 + \eta_2}$$

$$= \frac{\frac{120\pi}{4} - 120\pi}{120\pi + \frac{120\pi}{4}}$$

$$\Gamma_E = \frac{\frac{1}{4} - 1}{1 + \frac{1}{4}} = -\frac{3}{5}$$

a) Fraction of power transmitted = $1 - |\Gamma_E|^2$

$$= 1 - \frac{9}{25} = \frac{16}{25}$$

$$\frac{P_t}{P_i} = \frac{16}{25}$$

b) Reflected $\vec{E}$ field fraction:

$$\frac{E_r}{E_i} = \Gamma_E = -\frac{3}{5}$$ (as calculated above)

c) Transmitted $\vec{H}$ field fraction:

$$\frac{\vec{H}_t}{\vec{H}_i} = 1 + (\Gamma_H) = 1 - \Gamma_E = 1 + \frac{3}{5}$$

$$\frac{\vec{H}_t}{\vec{H}_i} = \frac{8}{5}$$

Previous Years' Question

A uniform plane wave travelling in air is incident on the plane boundary between air and another dielectric medium with $\epsilon_r = 4$. The reflection coefficient for the normal incidence, is

a) zero **b)** $0.5\angle 0°$

c) $0.333\angle 180°$ **d)** $0.333\angle 90°$

Reflection at perfect conductor

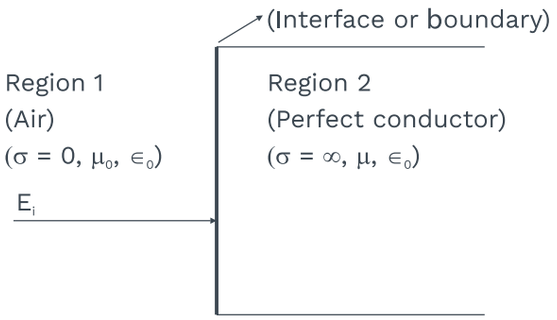


Fig. 5.9 Reflection at Perfect Conductor

1) For perfect conductor

$\sigma \to \infty$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}} = 0$$

Hence, for normal incidence:

$$\Gamma_E = \frac{\eta_2 - \eta_1}{\eta_1 + \eta_2}$$

$$\Gamma_E = \frac{0 - 120\pi}{0 + 120\pi}$$

$$\Gamma_E = -1$$

$$\Gamma_H = -\Gamma_E = 1$$

and

$$\Gamma_P = -1$$

This implies that there is complete reflection at conductor surface.

As

$$\Gamma_E = -1$$

$$\tau_E = 1 + \Gamma_E = 1 - 1$$

$$\tau_E = 0$$

That is, zero transmission of wave into perfect conducting medium.

$$E_i + E_r = E_t = 0$$

and

$$\tau_H = 1 + \Gamma_H = 1 + 1 = 2$$

and

$$H_i + H_r = H_t = H_{max}$$

Rack Your Brain

The incident wave amplitude is 17units. Find the reflected wave amplitude if the transmission coefficient is 0.6.

2) when Γ_E is very close to –1, Γ_H is very close to +1.

3) Γ_E being negative means the reflected electric field is changing its direction and hence cancels with the incident electric field. Due to this the electric field near the boundary is zero, and the magnetic field is maximum.

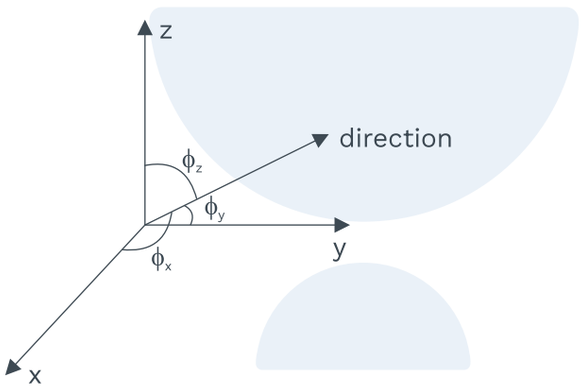
Oblique-incidence/Inclined incidence

The general form of a uniform plane wave in any arbitrary direction is given by:

$$\vec{E} = \vec{E}_0 e^{j(\omega t - \vec{k}\cdot\vec{R})} \qquad ...(1)$$

where

$\vec{k} = \beta_x \hat{x} + \beta_y \hat{y} + \beta_z \hat{z}$ = Propagation vector



And

$\vec{R} = x\hat{x} + y\hat{y} + z\hat{z}$ = Position vector

$\beta_x = \beta\cos\phi_x$
$\beta_y = \beta\cos\phi_y$
$\beta_z = \beta\cos\phi_z$

Rack Your Brain

What is the phase velocity of a wave in a perfect conductor?

ϕ_x, ϕ_y and ϕ_z are the angle between the direction of propagation and axis x, y and z, respectively.

Now, equation 1) can be written as:

$$\vec{E} = \vec{E}_0 e^{j(\omega t - \beta_x x - \beta_y y - \beta_z z)}$$

This form is known as phase constant form, and in this phase constant β is given by the magnitude of propagation vector $\vec{k}$, that is,

$$\beta = |\vec{k}| = \sqrt{\beta_x^2 + \beta_y^2 + \beta_z^2}$$

Reflection at inclined plane

In this section, we are going to discuss a more general case. Consider the oblique incidence of the plane wave as shown in the below figure.

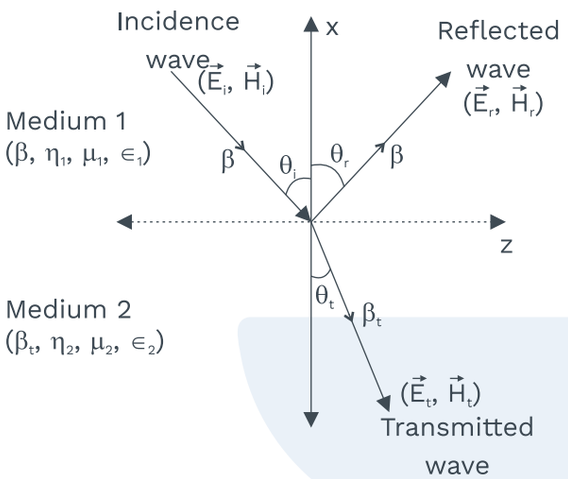


Fig. 5.10 Oblique Incidence of a Plane Wave

- The incidence plane wave $(\vec{E}_i, \vec{H}_i)$ is incidence on the interface at an angle θ_i with normal. Some portion of the incident wave is reflected back into the same medium at an angle of θ_r and the rest of the portion is transmitted to the second medium at an angle of θ_t.
- Here y-z plane at x = 0 represents the boundary and propagation of the wave in z-x plane.
- Direction of propagation = $\beta_x a_x + \beta_z a_z$.

Snell's law

Snell's law gives the relation between angle of incidence and angle of refraction:

i) $\theta_i = \theta_r$ (angle of incidence = angle of reflection)

ii) $\frac{\sin\theta_i}{\sin\theta_t} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$

Previous Years' Question

In a good conductor the phase relation between the tangential components of electric field E_t and the magnetic field H_t is as follows:

a) E_t and H_t are in phase
b) E_t and H_t are out of phase
c) H_t leads E_t by 90°
d) E_t leads H_t by 45°

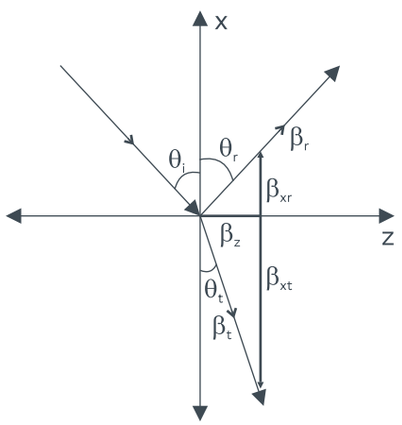


Fig. 5.11 Snell's Law

From the above figure:

$$\beta_z = \beta_t \cos(90° - \theta_t)$$

$$= \beta_r \cos(90° - \theta_r)$$

$$\beta_t \sin\theta_t = \beta_i \sin\theta_i$$

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{\beta_t}{\beta_i} = \frac{\omega\sqrt{\mu_2 \epsilon_2}}{\omega\sqrt{\mu_1 \epsilon_1}} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$$

Now, for the dielectric medium:

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{V_1}{V_2} = \frac{\omega\sqrt{\mu_0 \epsilon_2}}{\omega\sqrt{\mu_0 \epsilon_1}} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$\Rightarrow$ If $\theta_t = 90°$,
then $\theta_i = \theta_c$ = critical angle
and
For all $\theta_i > \theta_c$
= zero transmission for wave in medium 2.
$\Rightarrow$ For total internal reflection

$$\Rightarrow \quad \frac{\sin\theta_c}{\sin 90°} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$\frac{\sin\theta_c}{1} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$\epsilon_1 > \epsilon_2$

That is, wave must travel from denser to rarer medium.

Rack Your Brain

As the wave travels across a boundary into a new medium which characteristics of the wave would not change?

Grey Matter Alert!

Magnetic field expression for uniform plane wave travelling in arbitrary direction is

$$\vec{H} = \frac{1}{\omega\mu}\vec{k} \times \vec{E}$$

where $\vec{k}$ = propagation vector

Example 16:
What is the minimum angle for total internal reflection for a wave propagating from a dielectric medium $\epsilon_r = \sqrt{3}$ into free space?

Sol:
Minimum angle for total internal reflection
= Critical angle
By Snell's law $n_1 \sin\theta_i = n_2 \sin\theta_t$
For finding critical angle
$n_1 \sin\theta_i = n_2 \sin 90°$

$$\sin\theta_i = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_{r_2}}{\epsilon_{r_1}}} = \sqrt{\frac{1}{\sqrt{3}}}$$

$$\theta_i = \sin^{-1}\sqrt{\frac{1}{\sqrt{3}}} = 49.45°$$

Rack Your Brain

What is the difference between uniform plane wave and plane wave?

SURFACE AND PLANE POLARISATION (S AND P POLARISATION)

Based on general introduction of oblique incidence, we will now consider two, case:

i) S (surface) polarised wave: In this electric field $\vec{E}$ is parallel to boundary of surface. It is also known perpendicular polarisation.

ii) P (plane) polarised wave: Electric field $\vec{E}$ direction is in same plane as wave propagation. It is also known as parallel polarisation.

The incident, reflected and transmitted electric fields are in the same plane where the x-z plane is used for the propagation and y–z plane is used as boundary plane as shown in the below figure:

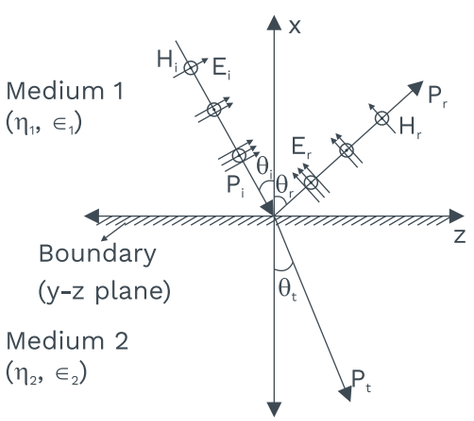


Fig. 5.12 Reflection and Transmission

For S(Surface) polarised wave:

$\vec{E} = E(x, z, t)_y$

$\vec{H} = E(x, z, t)_{(x,z)}$

Here $\vec{E}_y$ is parallel to boundary (y–z) plane.

For P (Plane) polarised wave:

$\vec{E} = E(x, z, t)_{(x,z)}$

$\vec{H} = E(x, z, t)_y$

Here electric field direction is in same plane as propagation.

Now, for S-polarised wave:

$E_i + E_r = E_t$ ...(1)

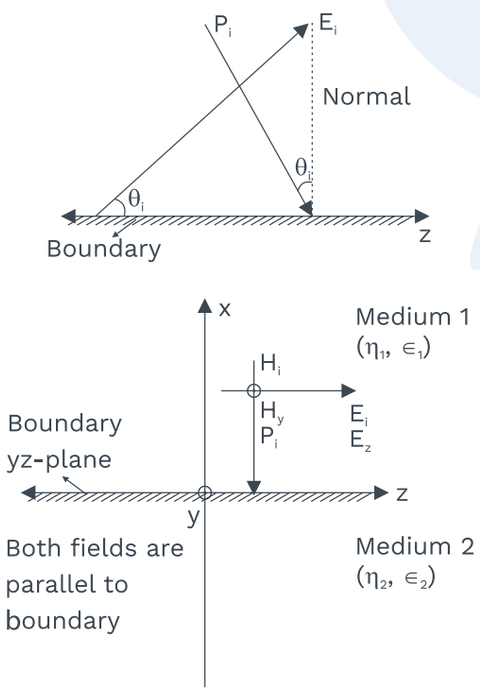


Fig. 5.13 Perpendicular and Parallel Polarization

For P-polarised wave

$E_{t1} = E_{t2}$

$E_i \cos\theta_i + E_r \cos\theta_r = E_t \cos\theta_t$

$H_{t1} = H_{t2}$

Previous Years' Question

For an electromagnetic wave incident form one medium to a second medium, total reflection takes place when:

a) The angle of incidence is equal to the Brewster angle with E field perpendicular to the plane of incidence.

b) The angle of incidence is equal to the Brewster angle with E field parallel to the plane of incidence.

c) The angle of incidence is equal to the critical angle with the wave moving from the denser medium to a rarer medium.

d) The angle of incidence is equal to the critical angle with the wave moving from a rarer medium to a denser medium.

Grey Matter Alert!

Concept of total internal reflection is valid for both perpendicularly and parallelly polarised EM wave. But the phase shift undergone by reflected wave is different for different polarisation.

$H_i + H_r = H_t$

Using $E_i = \eta_1 H_i$, $E_t = \eta_2 H_t$, $E_r = \eta_1 H_r$

Above two equation from $\vec{E}$ and $\vec{H}$ used for solving for Γ_P and Γ_S

$$\Gamma_P = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$$

Similarly:

$$\Gamma_s = \frac{\frac{\eta_2}{\cos\theta_t} - \frac{\eta_1}{\cos\theta_i}}{\frac{\eta_2}{\cos\theta_t} + \frac{\eta_1}{\cos\theta_i}}$$

or

$$\Gamma_s = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

Now for normal incidence:

$\theta_i = \theta_r = \theta_t = 0$

$$\Gamma_s = \Gamma_P = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

- In general incidence

$\Gamma_s \neq \Gamma_P$

Brewster's angle

Brewster angle is an angle of incidence at which light with a particular polarisation is perfectly transmitted through a transparent dielectric surface with no reflection.

That is, at Brewster's angle:

Reflection coefficient, $\Gamma = 0$

Transmission coefficient, $\tau = 1$

- Concept of Brewster's angle is not valid for S-polarised (perpendicular polarised) EM wave for dielectric media interface.

That is $\theta_{B\text{-}s}$ = Does not exist

and $\theta_{B-P} = \tan^{-1}\sqrt{\frac{\epsilon_2}{\epsilon_1}}$

Previous Years' Question

Refractive index of glass is 1.5. Find the wavelength of a beam of light with frequency of 10^{14} Hz in glass. Assume velocity of light is 3×10^8 m/s in vacuum:

a) 3 μm **b)** 3 mm

c) 2 mm **d)** 1 mm

Rack Your Brain

How the velocity of EM wave changes when it moves to a different medium?

Rack Your Brain

If any wave is incident on a dielectric at Brewster's angle then the polarisation of reflected wave is _____.

Example 17:
Calculate the Brewster's angle of a parallel polarised wave travelling from air into glass for which $\in_r = 6$.

Sol:
For parallel polarisation,

Brewster's angle $\theta_B = \tan^{-1}\sqrt{\frac{\in_{r_2}}{\in_{r_1}}}$

$$\theta_B = \tan^{-1}\sqrt{6}$$

$$\theta_B = 67.79°$$

Brewster's angle = $\theta_B = 67.79°$

- EM wave are composed of oscillating magnetic field and electric fields.
- In static fields, the electric and the magnetic fields are the functions of space only while in time-varying field, the electric and the magnetic field are functions of space and time both.
- **Faraday's law :** EMF or potential difference can be induced in any closed loop when the magnetic flux changes with time.

$$V_{emf} = -N\frac{d\psi_m}{dt}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\psi_m}{dt}$$

where E = Electric field,
ψ = Flux through the closed loop and
N = Number of turns

- Conduction current very large in comparison of displacement current. That is, $J_c >> J_d$

Chapter Summary



- Displacement current density, $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon \frac{\partial \vec{E}}{\partial t}$
- Conduction current, $J_c = \sigma E$
- Maxwell's equations:

Differential form	Integral form
$\nabla \cdot \vec{D} = \rho_v$	$\oint \vec{D} \cdot ds = \int_v \rho_v dV$
$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot ds = 0$
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_s \vec{B} \cdot d\vec{s}$
$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint \vec{H} \cdot d\vec{l} = \int_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$

- Helmholtz's equation of wave in free space:

  $\nabla^2 \vec{E}_s = j\omega\mu(\sigma + j\omega\epsilon)\vec{E}_s$

  $\nabla^2 \vec{H}_s = j\omega\mu(\sigma + j\omega\epsilon)\vec{H}_s$

  or

  $\frac{d^2 \vec{E}_x}{dz^2} = \gamma^2 \vec{E}_x$

  $\frac{d^2 \vec{H}_x}{dz^2} = \gamma^2 \vec{H}_y$
- Propagation constant $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$,

  where α = attenuation constant, β = phase constant.
- Every EM wave is a time and space harmonic field advancement with transverse (orthogonal) $\vec{E}$ and $\vec{H}$ wave propagation.

  $E(z, t)_x = E_0 e^{j\omega t} e^{-\gamma z} \hat{a}_x$

  $H(z, t)_y = H_0 e^{j\omega t} e^{-\gamma z} \hat{a}_y$

  As $\gamma = \alpha + j\beta$

  $E(z, t)_x = H_0 e^{-\alpha z} e^{j(\omega t - \beta z)} \hat{a}_x$

  $H(z, t)_y = H_0 e^{-\alpha z} e^{j(\omega t - \beta z)} \hat{a}_y$

- **Intrinsic impedance:** The ratio of electric field to magnetic fields present in any medium.

$$\eta = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\gamma}{\sigma + j\omega \in}$$

$$\eta = \sqrt{\frac{j\omega\mu}{(\sigma + j\omega \in)}}$$

- Magnitude of intrinsic impedance indicate the magnitude ratio of electric field to the magnetic field.
- Angle of intrinsic impedance indicates the phase difference between electric and magnetic field.
- Intrinsic impedance for free space $= \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\,\pi.$
- Intrinsic impedance for dielectrics $= 120\pi\sqrt{\frac{\mu_r}{\epsilon_r}}.$
- Loss tangent $= \frac{|\vec{J}_c|}{|\vec{J}_d|} = \frac{\sigma}{\omega \in} = \tan\delta$ and δ = loss angle.
- Loss angle = $\delta = \tan^{-1}\left(\frac{\sigma}{\omega \in}\right)$
- Characterization of medium:

  i) Loss-less medium $\rightarrow \frac{\sigma}{\omega \in} = 0$

  ii) Lossy medium $\rightarrow \frac{\sigma}{\omega \in} \neq 0$

  a) Low-loss medium $\frac{\sigma}{\omega \in} < 1$

  b) High-loss medium $\frac{\sigma}{\omega \in} >> 1$

- For loss-less medium ($\sigma = 0$, $\mu = \mu_r\mu_0$, $\in = \in_0\in_r$) $\left(\frac{\sigma}{\omega \in} = 0\right)$

  Propagation constant, $\gamma = \sigma + j\beta = j\omega\sqrt{\mu \in}$

  Attenuation constant, $\alpha = 0$

  Phase constant, $\beta = \omega\sqrt{\mu \in} = \frac{2\pi}{\lambda}$

Velocity, $v = \frac{\omega}{\beta}$

Intrinsic impedance, $\eta = \sqrt{\frac{\mu}{\epsilon}}$

- For lossy medium ($\sigma \neq 0$, $\mu = \mu_0\mu_r$, $\epsilon = \epsilon_r\epsilon_0$)

  Propagation constant, $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$

  Attenuation constant, $\alpha = \omega\sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1\right]}$

  Phase constant, $\beta = \omega\sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1\right]}$

- For loss-less medium $\left(\frac{\sigma}{\omega\epsilon} < 1\right)$

  Attenuation constant, $\alpha = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$

  Phase constant, $\beta = \omega\sqrt{\mu\epsilon}$

  Velocity, $v = \frac{1}{\sqrt{\mu\epsilon}}$

  Impedance, $\eta = \sqrt{\frac{\mu}{\epsilon}}$ (real)

- For high-loss medium $\left(\frac{\sigma}{\omega\epsilon} >> 1\right)$

  Propagation constant, $\gamma = \sqrt{\frac{\omega\mu\sigma}{2}} + j\sqrt{\frac{\omega\mu\sigma}{2}}$

  Attenuation constant, $\alpha = \sqrt{\frac{\omega\mu\sigma}{2}}$

  Phase constant, $\beta = \sqrt{\frac{\omega\mu\sigma}{2}}$

  Velocity, $v = \sqrt{\frac{2\omega}{\mu\sigma}}$

Intrinsic impedance, $\eta = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45°$

$$\eta = \sqrt{\frac{\omega\mu}{2\sigma}} + j\sqrt{\frac{\omega\mu}{2\sigma}} = R_s + jX_s$$

- **Skin depth or depth of penetration:** It is defined as the distance over which plane wave is attenuated by a factor of e^{-1} in a lossy medium.

  Skin depth, $\delta = \frac{1}{\alpha} = \frac{1}{\beta} = \sqrt{\frac{2}{\omega\mu\sigma}}$

  $\delta = \frac{1}{\sigma R_s} = \frac{1}{\sigma X_s}$

  R_s = Real part of intrinsic impedance
  X_s = Imaginary part of intrinsic impedance
- Velocity and skin depth relation:
  Velocity = $\omega\delta$
- Wavelength and skin-depth relation:
  $\lambda = 2\pi\delta$
- Power density is the amount of power per unit volume. It is calculated using power flow in the EM field.

  $$\oiint (\vec{E} \times \vec{H}) = -\frac{d}{dt} \iiint \left[\frac{1}{2} \in E^2 + \frac{1}{2} \mu H^2 \right] dV - \iiint \sigma \vec{E}^2 dV$$
- **Poynting theorem:** The net power flowing out of any given volume V is equal to the time rate of decrease in the energy stored within the given volume V minus the ohmic losses.
- Poynting vector = $\vec{E} \times \vec{H}$ = it signifies the EM wave power density, which is crossing the surface in propagation direction.
- Time-average power density $\vec{P}_{avg} = \frac{1}{2} Re(\vec{E} \times \vec{H})$

  $$= \frac{1}{2\eta} |\vec{E}|^2$$

  $$= \frac{|\vec{H}|^2 \eta}{2}$$
- Total average power crossing a given surface s is:

  $$P_{avg} = \int_s \vec{P}_{avg} \cdot d\vec{s}$$

- Polarisation is defined as electric fields orientation and the possible planner components satisfying the basic transverse nature. It is the study of time behavior of EM-wave or it defines the locus drawn by electric field vector as wave propagates.
- Condition for linear polarisation: Phase difference between different components of electric field is $n\pi$, i.e. $\Delta\phi = \pm n\pi$
- Condition for circular polarisation:

  i) $\left|\vec{E}_x\right| = \left|\vec{E}_y\right|$

  That is, amplitude of different components of electric field are equal.

  ii) Phase difference, $\Delta\phi = (2n+1)\frac{\pi}{2}$ →Left circularly polarised or clockwise.

  Phase difference, $\Delta\phi = -(2n+1)\frac{\pi}{2}$ →Right circularly polarised or counter clockwise.
- Condition for elliptical polarisation:

  $\left|\vec{E}_x\right| \neq \left|\vec{E}_y\right|$

  $\Delta\phi = (2n+1)\frac{\pi}{2}$ →Left elliptically polarised

  $\Delta\phi = -(2n+1)\frac{\pi}{2}$ →Right elliptically polarised
- General case, EM wave is elliptically polarised if $\Delta\phi \neq 0$, $\Delta\phi \neq \frac{\pi}{2}$
- Reflection coefficient:

  $\Gamma_E = \frac{E_r}{E_i}$

  $\Gamma_H = \frac{H_r}{H_i}$
- Transmission coefficient:

  $\tau_E = \frac{E_t}{E_i}$

  $\tau_E = 1 + \Gamma_E$

- For normal incidence:

$$\Gamma_E = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau_E = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\Gamma_H = -\Gamma_E = \frac{\eta_1 - \eta_2}{\eta_2 + \eta_1}$$

- Reflection of perfect conductor:
  $\Gamma_E = -1$
  $\Gamma_H = -\Gamma_E = 1$
  $\tau_E = 1 + \Gamma_E = 1 - 1 = 0$
  $\tau_E = 0$
  This implies that there is complete reflection at conductor surface, i.e. zero transmission of wave into a perfect conducting medium.
- The electric field near the conductor boundary is zero and the magnetic field is maximum.
- Uniform plane in any arbitrary direction
  $\vec{E} = \vec{E}_0 e^{j(\omega t - \vec{k}\cdot\vec{R})}$

  $\vec{k} = \beta_x \hat{x} + \beta_y \hat{y} + \beta_z \hat{z}$ = Propagation vector

  $\vec{R} = x\hat{x} + y\hat{y} + z\hat{z}$ = Position vector

  $\beta = \sqrt{\beta_x^2 + \beta_y^2 + \beta_z^2} = |\vec{k}|$
- **Snell's law:** It gives the relationship between the angle of incidence and the angle of refraction.
  i) $\theta_i = \theta_t$
  ii) $\frac{\sin\theta_i}{\sin\theta_t} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$

  or

  $\beta_t \sin\theta_t = \beta_i \sin\theta_i$
- For dielectric medium, $\frac{\sin\theta_i}{\sin\theta_t} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$

- Critical angle: The angle of incidence that provides an angle of refraction of 90°.

$$\sin\theta_c = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$\theta_c = \sin^{-1}\sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

- Brewster's angle is an angle of incidence at which light with a particular polarisation is perfectly transmitted through transparent dielectric surface with no reflection.
  That is, $\Gamma = 0$ or $\tau = 1$
- Brewster's angle for dielectric medium.

  $\theta_{B\text{-}s}$ = Does not exist or $\theta_{B-p} = \tan^{-1}\sqrt{\frac{\epsilon_2}{\epsilon_1}}$

PRACTICE QUESTIONS

Q1 Which of the following is a Maxwell's equation?

a) $\nabla \times \vec{H} = 0$ b) $\nabla \cdot \vec{B} = \mu_0 \vec{J} + \mu_0 \frac{\partial \vec{D}}{\partial t}$

c) $\nabla \cdot \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ d) $\nabla \cdot \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

Q2 Which of the following Maxwell's equation represents modified Ampere's law?

a) $\nabla \cdot \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ b) $\nabla \cdot \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ c) $\nabla \cdot \vec{D} = 0$ d) $\nabla \vec{D} = \rho_v$

Q3 Which of the following law is not represented by Maxwell's equation?

a) Maxwell's law b) Ampere's law

c) Planck's law d) None of these

Q4 In electromagnetic wave, the phase difference between component of EM wave and propagation of EM wave is:

a) zero b) $\frac{p}{2}$ c) $\frac{3p}{2}$ d) π

Q5 Calculate the value of intrinsic impedance in a medium.
(Given $\mu = 3\mu_0$, $\in = 12\in_0$, $\sigma = 0$)

a) 120π b) 60π c) 40π d) 30π

Q6 Skin depth in a medium for a EM wave at 1 MHz is 4 μm. Calculate the skin depth for the same medium at 16 MHz.

a) 1 μm b) 2 μm c) 0.25 μm d) 4 μm

Q7 **Which of the following is condition for high-loss medium? (δ is loss angle)**

a) $\tan\delta = 0$ **b)** $\tan\delta >> 1$ **c)** $\tan\delta << 1$ **d)** $\tan\delta = 1$

Q8 **In a free space the electric field is given by** $\vec{E} = 10\cos(\omega t - \beta z)\hat{a}_x$ V/ m**. Find the average power crossing a circular area of diameter 2 m in a plane z = 2.**

a) $\frac{50}{120}$ **b)** $\frac{100}{120}$ **c)** $\frac{150}{120}$ **d)** $\frac{120}{100}$

Q9 **The electric field vector of a plane wave is given by** $\vec{E} = (3\hat{a}_y + 4j\hat{a}_z)e^{j(\omega t - \beta x)}$ V/ m**. The polarisation of wave is**

a) Linear **b) Left circular** **c) Left elliptical** **d) Right elliptical**

Q10 **A wave is incident on a dielectric at Brewster's Angle. Then reflected wave is always:**

a) Linear and perpendicular polarised
b) Linear and parallel polarised
c) Elliptically polarised
d) None of these

Answer key:

Q1	**c)**	**Q2**	**b)**	**Q3**	**c)**	**Q4**	**b)**	**Q5**	**b)**
Q6	**a)**	**Q7**	**b)**	**Q8**	**a)**	**Q9**	**b)**	**Q10**	**a)**